

Getting Your Hands Dirty With Deb Files



As a Linux user, you have to deal with packages occasionally, if not every day. This article talks about the Debian way of going about it!

First things first: a package, put simply, is software created especially for a particular GNU/Linux distribution. Debian packages are called 'deb' in short, and have the *.deb* file extension.

There are many advantages of using packages, rather than the upstream sources, to install any software. Packages are always signed by its maintainers. Some security updates come faster in a distribution compared to its upstream, since it affects the whole distribution rather than a single package.

Some internal details

Debian packages are handled by software called *dpkg*. Generally, users don't need to deal directly with *dpkg*. Its front-end is called *apt-get*, which takes care of almost all of your needs.

You can find all configuration files related to Apt package management in the */etc/apt/* directory. Let's look at each of the files:

- */etc/apt/sources.list* stores the list of repositories you are using. By default, this will be added, depending upon the Debian version you are using (stable, testing, unstable, etc). Make sure that you have security repositories enabled. You can comment, uncomment, add or delete any lines here, but don't forget to run *apt-get update* (see the package management section below) each time you edit this file.
- */etc/apt/apt.conf* is the general

configuration file where you can change many settings depending upon your need. If, for instance, you are behind a proxy, you can add the following line to your proxy settings:

```
Acquire: http://Proxy 'http://username:password@IP:Port';
```

Please see the man page of *apt.conf* for a detailed description.

Package management

Let's look at some practical examples. We know that *apt-get* is the front end to deal with Debian packages:

```
apt-get update # to update the internal package database
apt-get install package_name
```

Now, if you are not sure about the exact *package_name*, you can search in your package database:

```
apt-cache search package_name
```

You can also view more information about packages, using the following command:

```
apt-cache show package_name
```

This will display the package description, package size, maintainer information, etc, to help you decide if this is the package that you're looking for. Removing a package is also very easy: